

Tan Jun Xian

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Education

University of Texas, Austin <i>Master of Science, Artificial Intelligence</i>	Jan 2024 - Apr 2025
Nanyang Technological University <i>Bachelor of Engineering, Computer Science, Artificial Intelligence</i>	Aug 2019 - Dec 2022
Temasek Polytechnic <i>Diploma in Computer Engineering, Robotics</i> <i>Diploma Plus in Psychology</i>	Apr 2014 - May 2017

Experience

IOTA MedTech <i>AI & Software Engineer</i> – Architected and developed a full-stack clinical decision support platform enabling medical professionals to query complex data through a conversational interface. – Engineered a Retrieval-Augmented Generation (RAG) pipeline to ground an LLM in a proprietary medical knowledge base, using extensive prompt instruction tuning to optimize for clinical accuracy, with added safety guardrails for privacy. – Led the full software development lifecycle for a medical imaging device, securing regulatory approval by ensuring ISO 13485 compliance. – Deployed scalable, cloud-native AI solutions on AWS (ECS, S3, RDS), optimizing for cost and performance while adhering to regulatory and security requirements.	Jan 2023 — Mar 2026
Panasonic R&D Center <i>AI Engineer Intern</i> – Collaborated with developers and researchers to create AI solutions tailored for external clients, focusing on data acquisition, preprocessing, and model training – Enhanced and fine-tuned deep learning architectures to optimize performance, contributing to robust AI model development – Engineered a deep learning web application using ReactJS, NodeJS/ExpressJS, and MySQL to provide scalable AI solutions – Documented and presented experimental findings and analysis outcomes to senior engineering stakeholders for review and feedback	Jul 2022 — Dec 2022
Trendlines Medical <i>Machine/Deep learning Engineer Intern</i> – Conducted AI software development for medical image analysis utilizing PyQT framework – Developed and optimized deep learning models for image stitching, super resolution, and 3D scene reconstruction – Conducted time-series analysis employing machine learning algorithms, coupled with proficient data visualization techniques	May 2022 — Jul 2022
Temasek Polytechnic <i>Software Engineer (Robotics)</i> – Developed software for an autonomous robot	Feb 2016 – Mar 2017

Technical Skills

Cloud AWS, GCloud
Languages Python, Bash, Java, C, C++
OS Ubuntu, WSL2, AWS AMI

AI/ML Vision – Classification, Segmentation, Detection
AI/ML Text – LLMs, Summarization, Rec System
AI/ML Audio – ASR

Certificates

- ISO 13485:2016 Standard, 21 CFR Part 820 and 21 CFR Part 11 Regulation Training Course
- ISO 13485:2016 Internal Auditor Training Course
- Project Management Professional (PMP) (SF)
- CITI Program (Human, Data or Specimens Only Research)